

Multimedia related

1. Analysis of Huaping problem

1.1. Determine whether the ISP output is normal

```
/ # cat /sys/class/mstar/isp0/isp ints
frame_count          : 356229
frame interval       : 33330 us
P0_FIFO_FULL        : 0
P1_FIFO_FULL        : 0
P0_VS_ERR           : 0
P1_VS_ERR           : 0
ISPIF_VSYNC         : 356229, 71.492671, 0
AE_DONE             : 356228, 71.491522, 1074
AE_DONE period      : 32181 us
AWB_DONE            : 356228
AF_DONE            : 356228
AF_DMA_ERR          : 0
MLOAD_DONE          : 63
SW_INT_OUTPUT_DONE  : 356228, 71.491754
OUTPUT_DONE period  : 32413 us
ISP_IDLE            : 356229, 71.491753
ISP_BUSY period     : 32412 us
RDMA0_DONE          : 0
RDMA1_DONE          : 0
```

Under normal circumstances, the difference between frame cnt and Done Cnt is at most one. If there is a lot of accumulation, it means that the isp only receives vsync and no actual data.

```

~ # cat /sys/class/mstar/isp0/isp_ints
frame_count                : 46646
frame_interval             : 50099 us
P0_FIFO_FULL              : 0
P1_FIFO_FULL              : 0
P0_VS_ERR                 : 0
P1_VS_ERR                 : 0
ISPIF_VSYNC               : 46646, 07.704367, 0
AE_DONE                   : 35532, 07.643162, 2158
AE_DONE period            : 38907 us
AWB_DONE                   : 35532
AF_DONE                   : 35532
AF_DMA_ERR                : 0

```

As shown above.

If the isp output cnt is enough, then judge whether the isp output data content is normal through the scl pattern.

```
echo 2 > /sys/class/mstar/mscl/ptgen
```

pattern will replace the isp output data with pattern data. If the pattern is normal, it means that there is a problem with the isp output.

1.2. Determine whether the Vif output is normal

如果Vif 是frame mode

```
echo dump chnid portid /path > /proc/mi_modules/mi_vif/mi_vif0
```

如果是bayer 10/12 bit格式需要通过转换工具专成16bit 通过7 yuv观察画面。

如果vif这边也是整张frame 都是花的，怀疑vif 这边没有收到sensor 数据。

```

/ # /customer/riu_r 0x1304
BANK:0x1304
00: 0x8001 0x0000 0x0000 0x0000 0x0000 0x0000 0x1F00 0x0000
08: 0x0000 0x0000 0x0000 0x01DE 0x0000 0x0000 0x0052 0x0012
10: 0x8000 0x077F 0x0000 0x0437 0x0437 0x1FFF 0x0000 0x0000
18: 0x0001 0x0000 0x0000 0x0000 0x0780 0x0438 0x079C 0x0449
20: 0x0000 0x00FF 0x00FF 0x0202 0x4040 0x001F 0x001F 0x0000
28: 0x0000 0x0449 0x01E7 0x0000 0x0000 0x0000 0x0000 0x0000
30: 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x9000 0x007E
38: 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0003
40: 0x0001 0x0000 0x0000 0x0000 0x0000 0x0000 0x1300 0x0000
48: 0x0000 0x0000 0x0000 0x01FE 0x0000 0x0000 0x0012 0x0012
50: 0x0000 0x1FFF 0x0000 0x1FFF 0x1FFF 0x1FFF 0x0000 0x0000
58: 0x0001 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000
60: 0x0000 0x00FF 0x00FF 0x0202 0x4040 0x001F 0x001F 0x0000
68: 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000
70: 0x0000 0x0000 0x0000 0x0000 0x0000 0x9000 0x007E 0x0000
78: 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000 0x0000

```

h001c	h001c	12	0	reg_vif_ch0_crop_pix_cnt	12	0	13	r	VIF crop pixel count (to ISP)	VIF_CH0
h001d	h001d	12	0	reg_vif_ch0_crop_line_cnt	12	0	13	r	VIF crop line count (to ISP)	VIF_CH0
h001e	h001e	12	0	reg_vif_ch0_total_pix_cnt	12	0	13	r	VIF total pixel count (to ISP)	VIF_CH0
h001f	h001f	12	0	reg_vif_ch0_total_line_cnt	12	0	13	r	VIF total line count (to ISP)	VIF_CH0

如果这里出现有为0的情况说明vif 这边也没有收到data。

1.3. 判断sensor数据是否正常

```

echo devid > /sys/class/mstar/csi0/csi_dbg_mask
cat /sys/class/mstar/csi0/csi_ints

```

```
/ # cat /sys/class/mstar/csi0/csi_ints
FS_INTS          : 0
VC0_INTS         : 0
ERR_INT_DT       : 0
ERR_INT_PA_LENS  : 0
ERR_INT_PH_LENS  : 0
ERR_INT_ECC_ONEBIT : 0
ERR_INT_FRAME_START : 0
ERR_INT_FRAME_END : 0
ERR_INT_ECC_TWOBIT : 0
ERR_INT_CRC      : 0
ERR_INT_PA_WC_EQ0 : 0
ERR_INT_RAW10_LENS : 0
ERR_INT_CON_FE    : 0
ERR_INT_CON_FS    : 0
ERR_INT_LE        : 0
ERR_INT_LS        : 0
ERR_INT_OVERRUN   : 0
```

正常情况下都为0，这里反映数据协议是否有问题，如果有累加请sensor driver和硬件一起量一下信号质量或者clock是否有问题。

2. 常见的丢帧问题定位及解决方法

2.1. 常见丢帧原因

1. 应用调用MI_VENC_GetStream()取流不及时，比如设置30fps，每秒却只调用20次取流接口。
 2. scl设置depth不够，output buffer轮循不过来，比如子码流设置20fps，depth却只设置1张。
 3. 硬件Scale或者venc工作异常，出现Rfull或者venc hung导致丢了几帧数据。
 4. 应用调用MI_VENC_SetSuperFrameCfg()开启超大帧丢帧或者重编机制，当出现size过大的帧会被丢弃。
-

2.2. 出现帧率不够或者断流时，定位过程

1. 留意串口和/proc/kmsg有没有异常打印 (mi_sys和mi_venc的debug_level不能设置小于2)

类似 `process exceed timeout MillsecAfterTaskEnqueued`

或者 `[MI WRN ]: _MI_SYS_Try_DequeueInputTaskNoLock[8674]:`

`[thread:venc0_P0_MAIN] module id[2] dev id[0] pass id[0] inputtask is not finished more than 2seconds 、 Rfull 、 IMI OFlow 、 [CMDQ]cmdq(1) ERR: WAIT_TRIG_TIMEOUT 。`

2. 如果多路流都有帧率不够的问题，查看sensor帧率是否足够：

```
/ # cat /sys/class/mstar/isp0/isp_ints
frame_count          : 49929
frame_interval       : 49985 us
ISP_FIFO_FULL        : 0
INT_PIPE1_FIFO_FULL  : 0
ISP_BUSY             : 0, 00.000000
ISP_IDLE             : 49870, 61.227331
AWB_DONE             : 49870
RDMA0_DONE           : 0
RDMA1_DONE           : 0
WDMA0_DONE           : 0
WDMA1_DONE           : 0
WDMA6_LINE_HIT       : 0
WDMA6_DONE           : 0
ISPIF_VSYNC          : 49870, 61.193836, 0
AE_DONE              : 49870, 61.226633, 1058
AE_DONE_period       : 32797 us
AE_WIN0_DONE         : 49870
SW_INT_OUTPUT_DONE   : 49870, 61.227329
OUTPUT_DONE_period   : 33493 us
RGBIR_HISTO_DONE     : 0
AWB_ROW_DONE         : 49870
HDR_ARRAY_HIT_COUNT  : 0
HDR_LINE_BUF_CTRL_DONE : 0
HDR_FRAME_DONE       : 0
3DNR0_WDMA_REF_DONE  : 0
3DNR0_WDMA_HIST_DONE : 0
3DNR0_NE_DONE        : 0
3DNR0_FRAME_DONE     : 49870, 61.227215
3DNR0_FRAME_DONE_period : 33379 us
ROT_FRAME_DONE       : 0
ROT_FRAME_DONE_period : 0 us
ROT_ERR_TRIG         : 0

ISPIF_VSYNC :49924, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

- a. Frame_interval: 略小于sensor帧率的倒数，并且保持稳定不会跳动很大。否则是sensor驱动问题。
- b. ISP_FIFO_FULL：表示isp统计的，从MIPI接收丢了几帧。正常应该为0，或者保持某个值不动。
- c. 如果ISP_FIFO_FULL值异常，大概率是isp出现了回档 (OSD叠加、scale异常、ring mode rfull和imi mode OFlow都有可能导致回档)。
- d. 如果ISP_FIFO_FULL值正常，Frame_interval和帧率不符合，可能是驱动问题，或者应用下给sensor帧率下错。

3. 检查sensor帧率设置：

```

/ # cat /proc/mi_modules/mi_sensor/mi_sensor0
-----Common info for mi_sensor0-----
ChnNum 1 EnChnNum 0 InPortNum 0 OutPortNum 1 CollectSize 0
-----Common info for mi_sensor0 only dump enabled chn-----
ChnId EnInPNum EnOutPNum MMAHeapName
-----Input port common info for mi_sensor0 only dump enabled port-----
ChnId PortId SrcFrmRate DstFrmRate RealSrcFrmRate user_buf_quota usrInjectQ_cnt BindInQ_cnt TotalPendingBuf_size usrLockedInjectQ_cnt
ChnId PortId newPulseQ_cnt nextToDoPulseQ_cnt curWorkingQ_cnt workingTask_cnt lazyRewindTask_cnt
ChnId PortId bind_module_id bind_module_name bind_chnid bind_PortId bind_Type bind_Param
ChnId PortId LowLatencyDelays LastStaticDelayAvems
-----Output port common info for mi_sensor0 only for enabled port-----
ChnId PortId usrDepth BufCntQuota usrLockedCnt totalOutPortInUsed DrvBkReffifoQ_cnt DrvBkReffifoQ_size
ChnId PortId SrcFrmRate DstFrmRate RealSrcFrmRate UsrGetFifoQ_cnt UsrGetFifoQ_size UsrGetFifoQ_seqnum UsrGetFifoQ_discardnum
ChnId PortId workingTask_cnt FinishedTask_cnt
-----BindPeerInputPortList-----
ChnId PortId bind_module_id bind_module_name bind_chnid bind_PortId
-----start dump Pad info-----
PadId PlaneMode benable bmirror bflip tps ResCnt intfmode hdrmode planeCnt
0 0 0 0 0 20 1 1 2
MipiAttr YuvOrder 0 HdrHwMode 0 DataFmt 0 HdrVchNum 0 HsyncMode 0 LaneNum 0 LPackType0 LPackType1 samDelay 0
Res strResDesc cropX CropY CropW CropH OutW OutH MaxFps MinFps
Cur 1920x1080@30fps 0 0 1920 1080 1948 1097 30 3
-----End dump Pad info-----
-----start dump plane info-----
PadId PlaneId SnrName BayerId ePixPrec eHdrSrc CropX CropY CropW CropH
0 0 0 IMX307_MIPI RG 128PP 0 0 0 1920 1080
0 0 0 IMX307_MIPI RG 128PP 0 0 0 1920 1080
/ #

```

4. 如果是单路流帧率不够，其他路流帧率正常，检查scl output framerate 和 venc output framerate

```

/ # cat /proc/mi_modules/mi_scl/mi_scl0
-----Common info for mi_scl-----
ChnNum 34 EnChnNum 1 PassNum 1 InPortNum 1 OutPortNum 6 CollectSize 0
-----CMDQ Kickoff counter-----
PassId CmdqId TotalKickoff KickoffFence Idle
0 1 1522 1016 1012
DevId current_buf_size Peak_buf_size
0 0 0
each dev buf info:
offset length used_flag task_name
-----Common info for mi_scl only dump enabled chn-----
ChnId PassNum EnInPNum EnOutPNum MMAHeapName
0 0 1 2 (null)
ChnId current_buf_size Peak_buf_size user_pid
0 0 5fa000 178000 1025
each chn buf info:
offset length used_flag task_name
1f61000 2fd000 1 scl0-out0-0
225e000 2fd000 1 scl0-out0-1
-----chn0 pass0 pipeline delay in us-----
Flow Min Max Avg Diff
OnPreProcessInputTask 51601 174735 67353 66811
EnqueueInputTask 52229 175021 67731 67193
BarrierInputTask 68275 192510 69592 68521
CheckInputTaskStatus 99986 234408 100872 100002
DequeueInputTask 99991 234412 100876 100006
OnPollingAsyncOutputTaskConfig -1 0 0 0
EnqueueAsyncOutputTask -1 0 0 0
BarrierAsyncOutputTask -1 0 0 0
CheckOutputTaskStatus 100000 234434 100885 100016
DequeueOutputTask 100022 234455 100913 100037
FindMADispatch 100060 234537 100954 100075
-----Input port common info for mi_scl only dump enabled port-----
ChnId PassId PortId user_buf_quota UsrInjectQ_cnt BindInQ_cnt TotalPendingBuf_size usrLockedInjectQ_cnt
0 0 0 4 0 1 0 0
ChnId PassId PortId newPulseQ_cnt nextToDoPulseQ_cnt curWorkingQ_cnt workingTask_cnt lazyRewindTask_cnt
0 0 0 0 0 0 1 0
ChnId PassId PortId Enable bind_module_id bind_module_name bind_DevId bind_ChnId bind_PortId bind_Type bind_Param enable
0 0 0 1 33 mi_isp 0 0 0 4 0 1
ChnId PassId PortId SrcFrmRate DstFrmRate GetFrame/Ms FPS FinishCnt RewindCnt DropCnt
0 0 0 30/1 30/1 31/1033 30.00 507 0 0

```

```

-----Output port common info for mi_scl only for enabled port-----
ChnId  PortId  usrDepth  BufCntQuota  usrLockedCnt  totalOutPortInUsed  DrvBkReffifoQ_cnt  DrvBkReffifoQ_size
0 0 0 0 0 4 0 1 0 0
0 0 0 1 0 4 0 1 0 0
ChnId  PortId  UsrGetFifoCnt  UsrGetFifoQ_size  UsrGetFifoQ_seqnum  UsrGetFifoQ_discardnum
0 0 0 0 0 0
0 0 1 0 0 0
ChnId  PortId  workingTask_cnt  finishedTask_cnt
0 0 1 0
0 0 1 0
ChnId  PortId  GetFrame/Ms  FPS  FinishCnt  RewindCnt  GetTotalCnt  GetOkCnt
0 0 0 31/1033 30.00 507 0 508 508
0 0 1 31/1033 30.00 507 0 508 508

-----BindPeerInputPortList-----
ChnId  PortId  Enable  bind_module_id  bind_module_name  bind_Devid  bind_ChnId  bind_PortId  bind_Type  bind_Param  enable
0 0 0 1 0 mi_venc 0 0 0 1 0 1
0 0 1 1 0 mi_venc 0 0 0 1 0 1

-----start dump SCL module info-----
InitCnt  chip  version  UseSclMask
1 5 20x f

-----end dump SCL module info-----

-----end dump SCL Dev info-----
Devid  refcnt  source  cmdId  IrqEn  IrqMode  IrqNum  VsyncCnt  FrameDoneCnt  DropCnt  LowLatencyIsrCnt
0 1 0 0x f de587cb4 1 0 1 47 0 508 0 0, 0, 0
-----End dump SCL dev 0 info-----

-----start dump SCL CHN info-----
Devid  ChnId  start  Crop  rot  Size  pixel  Stride  Atom/Atom0
0 0 1( 0, 0, 1920, 1080) 0(1920, 1080) YUV422YUYV 0 1 505

Devid  ChnId  PreCnt  EngCnt  BarCnt  checkin  checkout  DeqCnt  DropCnt  EngQ0Null
0 0 508 508 508 507 507 507 0

-----End dump SCL CHN info-----

-----start dump SCL OUTPUT PORT info-----
Devid  ChnId  PortId  sclId  bindType  Enable  Pixel  bMirr/flip  PortCrop  OutputW  OutputH  Histogram  MatchLine  Stride  GetCnt  FailCnt  FinishCnt  fps
0 0 0 0 0 Frame 1 YUV420SP( 0, 0) 0, 0, 1920, 1080 1920 1080 1927 0 1920 508 0 507 29.99
0 0 1 1 1 Frame 1 YUV420SP( 0, 0) 0, 0, 1920, 1080 1920 1080 1927 0 1920 508 0 507 29.99
0 0 2 2 ERROR 0 YUV422YUYV( 0, 0) 0, 0, 0, 0 0 0 0 0 0 0 0 0.00
0 0 3 3 ERROR 0 YUV422YUYV( 0, 0) 0, 0, 0, 0 0 0 0 0 0 0 0 0.00
0 0 4 4 ERROR 0 YUV422YUYV( 0, 0) 0, 0, 0, 0 0 0 0 0 0 0 0 0.00
0 0 5 5 ERROR 0 YUV422YUYV( 0, 0) 0, 0, 0, 0 0 0 0 0 0 0 0 0.00
-----End dump SCL OUTPUT PORT info-----

```

BufCntQuota:应用设置的scl port buffer块数，如果是port1的fps不够，大概率是depth不够，output buffer轮循不过来。可以给port1的depth增加1块试试。

Fps: output port的实际输出帧率，如果不够，尝试该通道上所有RGN关闭。（RGN如果画太多，当它的带宽优化级较高时有可能出现scl port帧率不够）。

```

/ # cat /proc/mi_modules/mi_venc/mi_venc0
-----Common info for mi_venc0-----
ChnNum  EnChnNum  InPortNum  OutPortNum  CollectSize
16 1 0 1 0

-----Common info for mi_venc0 only dump enabled chn-----
ChnId  EnInPNum  EnOutPNum  MMAHeapName
0 1 0 (null)

-----Input port common info for mi_venc0 only dump enabled port-----
ChnId  PortId  SrcFrmRate  DstFrmRate  RealSrcFrmRate  user_buf_quota  usrInjectCnt  BindingCnt  TotalPendingBuf_size  usrLockedInjectCnt
0 0 0 20/1 20/1 0 0 2 3133440 0
ChnId  PortId  newPulseCnt  nextTodoPulseCnt  curworkingCnt  workingTaskCnt  lazyRewindTaskCnt
0 0 0 0 1 1 0

ChnId  PortId  bind_module_id  bind_module_name  bind_ChnId  bind_PortId  bind_Type  bind_Param
0 0 7 mi_vpe 0 1 16 1080

ChnId  PortId  LowLatencyDelayMs  LastStaticDelayAvGms

-----Output port common info for mi_venc0 only for enabled port-----
ChnId  PortId  usrDepth  BufCntQuota  usrLockedCnt  totalOutPortInUsed  DrvBkReffifoQ_cnt  DrvBkReffifoQ_size
ChnId  PortId  workingTask_cnt  finishedTask_cnt

-----BindPeerInputPortList-----
ChnId  PortId  bind_module_id  bind_module_name  bind_ChnId  bind_PortId

-----All VENC Dev info-----
Devid  CmdqId  CmdqSize  CmdqCnt  IrqNum  IsrTotalCnt  IsrFrmDoneCnt  IsrBufFullCnt  IsrRingFullCnt  IsrOtherCnt
0 0 0 0 0 26/27 620 619 1 0
Devid  Status  UtilHw  UtilMi  PeakHw  PeakMi  FPS  MBRate  %
0 2 0 0 0 0 20.00 163200 24

-----VENC 0 CHN info-----
ChnId  Devid  bStart  RefMemPA  RefMemSize  AlMemPA  AlMemVA  AlMemSize
0 0 0 1 0 3599360 2010000 c5130000 96928
ChnId  State  EnPred  base  enhance  MaxStreamCnt  FrameIdx  GardientH  GardientV  Fps_1s  kbps  Fps10s  kbps
0 2 0 0 1 0 3 1 11 20.00 2351 20.00 2000
ChnId  RingBufStartLine  RingBufHeight
0 0 1088

-----Input port of dev: 0-----
ChnId  width  Height  SrcFrmRate  MaxW  MaxH  FrameCnt  DropCnt  BlockCnt
0 1920 1080 20/1 1920 1080 620 0 0

-----Output port of dev: 0-----
ChnId  CODEC  Profile  BufSize  RefNum  bbyFrame  FrameCnt  DropCnt  ReEncnt  RingUnreadCnt  RingTotalCnt  usrLockedCnt
0 H264 0 1036800 0 1 619 0 0 0 0 0
ChnId  RateCtl  GOP  MaxBitrate  IPQPdelta  MaxQP  MinQP  MaxIQP  MinIQP
0 0 CBR 50 2000000 0 0 0 0 0 0

/ #

```

Inputport DropCnt:如果应用取流不及时，venc output buffer占满，encoder就开始drop yuv frame。因此这个值一直增加表示应用有取流不及时的行为。

OutputPort DropCnt:如果开启超大帧丢帧或者重编机制，或者encoder编出一帧size过大放不下Output buffer，该帧会被丢帧，这个值就会增加。

5. 举例

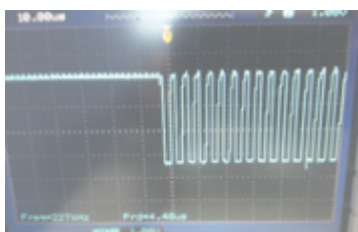
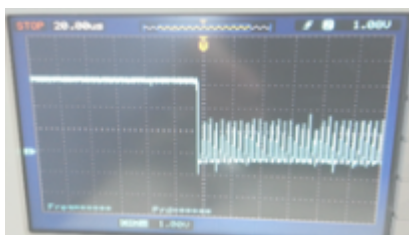
我们的venc ring buffer是长*宽*50%，里面最多能同时缓存MaxStreamCnt 默认这个值是6张。用户设置码率是6M。venc outputbuffer里偶尔出现要塞第6张的时候，剩余空间不够放第6张了，就会被丢掉，output dropCnt就会+1。解决output drop的方式是增加output buffer size，或者减少码率让每帧体积变小，或者减少MaxStreamCnt，但是减少MaxStreamCnt 意味用户取流要更快。

3. Sensor出图异常

3.1. I2C 不通

I2C不通时，请确认：

1. 工作电压是否正确
2. MCLK 是否正常
3. Power on 时序是否与spec 相同
4. 上拉电阻是否正常



(上拉不够)

3.2. MIPI CLK计算

format	resolution	frame rate	methodology	10-bit output MIPI data rate	DVP clock frequency
1080p	1920 x 1080	30 fps	full resolution qualified pixel (8+1920+8) x (8+1080+8)	420 Mbps/lane	84 MHz
qHD	960 x 540	60 fps	2x2 binning qualified pixel (4+960+4) x (4+540+4)	200 Mbps/lane	40 MHz
720p	1280 x 720	60 fps	crop from full resolution	375 Mbps/lane	75 MHz
VGA	640 x 480	60 fps	crop and binning	200 Mbps/lane	40 MHz

Spec是24MHZ mclk · 420Mbps/lane, MIPI信号进入HS mode · clock 上升沿跟下降沿都可以接收数据 · 若使用MCLK = 21.6M所以 $420 / 2 \times (21.6M / 24M) = 189M$

3.3. 画面紫色/红蓝颠倒

Bayer ID不对

Raw pixel format

- CUS_BAYER_RG



- CUS_BAYER_GR



- CUS_BAYER_BG



- CUS_BAYER_GB



3.4. MIPI garbage接收不到数据

1. LP->HS 的LP00过长

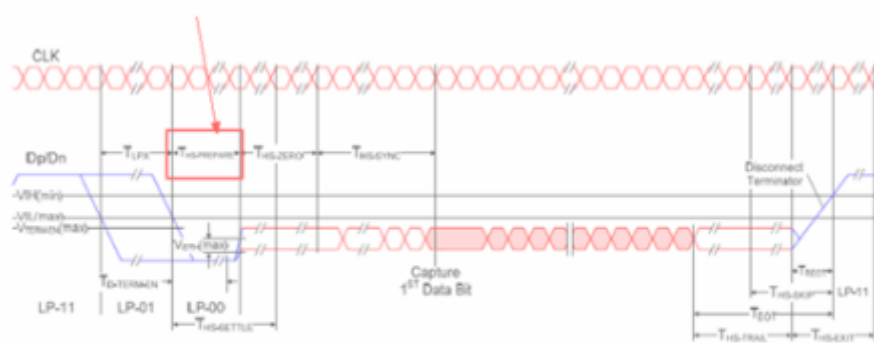
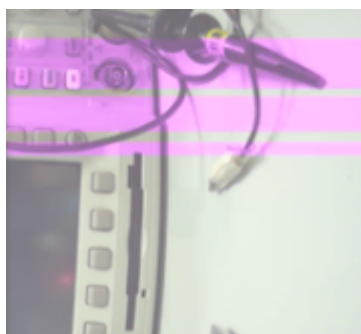


Figure 14 High-Speed Data Transmission in Bursts

2. MIPI swing: LP 1.2V, HS 200mv左右

3.5. Parallel sensor 粉色条状线

Connector 接触不良



3.6. CLK 不足导致sensor 数据接收不全

ISP clock > sensor clk

3.7. ISP 无法frame done

可能的原因如下：

1. 误判HS 信号
 2. Data 收不全/有瑕疵,加大window output size，再crop
 3. 频宽不够 FIFOfull
 4. 其他
-

3.8. 影像有dummy pixel/ ob data line



Image Data Output Format (CSI-2 output)

Frame Format

Each line of each image frame is output like the General Frame Format of CSI-2. The settings for each packet header are shown below.

DATA Type

Header [5:0]	Name	Setting register (P/C)	Description
00h	Frame Start Code	N/A	FS
01h	Frame End Code	N/A	FE
10h	NULL	N/A	Invalid data
12h	Embedded Data	N/A	Embedded data
20h	RAW10	Address: 41h, 42h (3441h, 3442h)	0A0Ah
2Ch	RAW12	CSI_DT_FMT [15:0]	0C0Ch
37h	OB Data	N/A	Vertical OB line data

Frame Structure

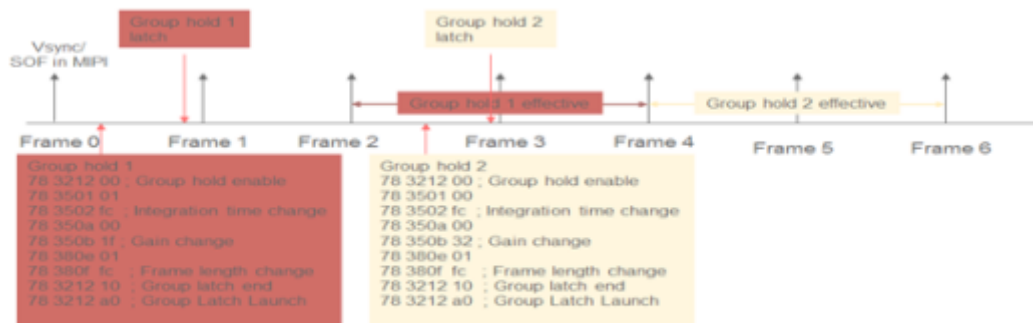


Frame Structure of CSI-2 output

3.9. Gain/shutter 生效不同步

Gain/shutter 要确保在同一个frame 生效必需使用group hold latch方法并确保 shutter/gain都在n+3帧生效。

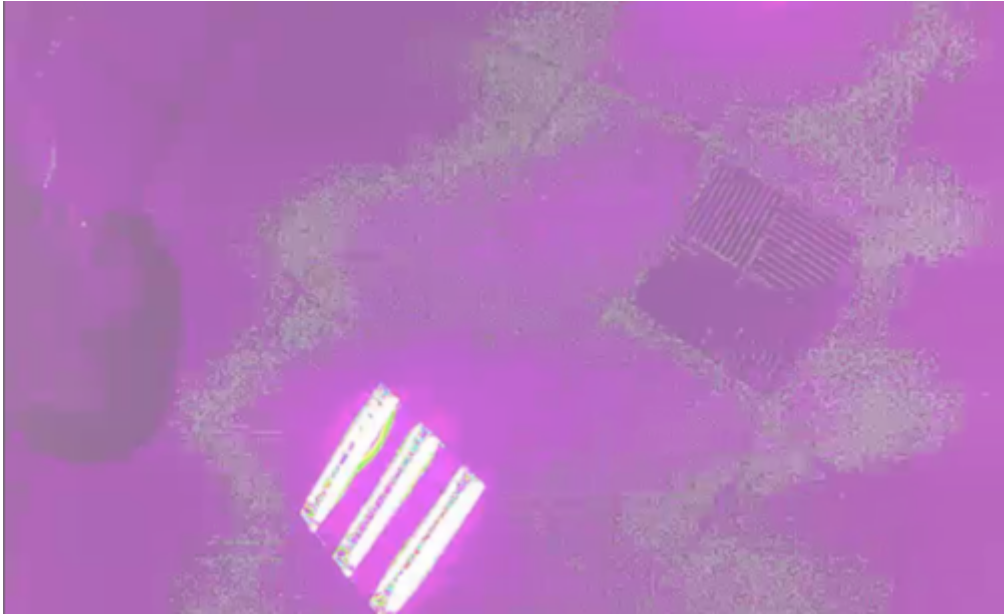
判断shutter/gain在第几帧生效可以设置manual AE，改变shutter/gain收AE log，看curY在第几帧变化。



This timing correspond to when there is one frame delay for gain (0x3503[5]=0)

3.10. PCLK 极性不对

32X系列只能改sensor极性，31X可以驱动改主控端



3.11. Parallel sensor Vsync / hsync 极性不对

数据接收不全

3.12. 高亮出现彩色

Sensor pad 设置不对，导致高位data bit 丢失



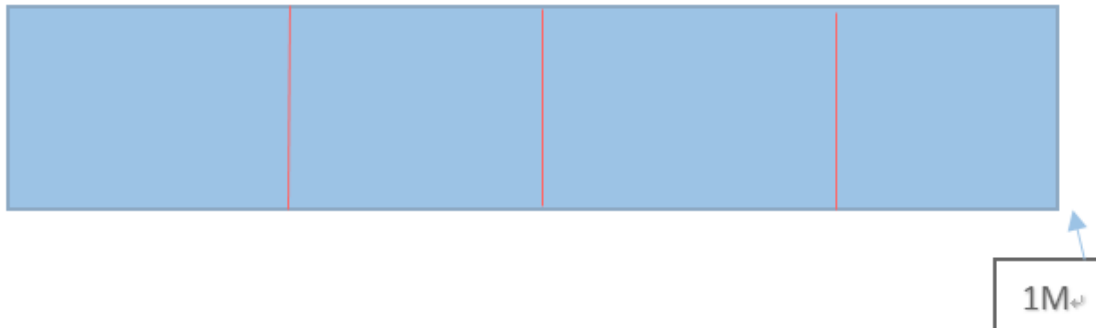
3.13. 高亮粉色

1. SC sensor 亮区发紫异常gain, Coarse gain (16'h3E08) bit[4:2],将低位置1,data|0x03
2. 高位溢出，改SENSOR_DATAPREC



4. venc output buffer的行为说明

以output buffer size(u32BufSize)为1M，u32MaxStrmCnt=4为例：



1. Create venc的时候，output buffer (venc-ring-mem) 即从mma里被申请好并且整块被映射成连续的虚拟地址。
2. Encoder从output buffer的起始位置填编码帧，假设放完第4帧，output buffer尾部还剩余空间，encoder也不会接着第5帧，而会从起始位置放第5帧。
3. 假设放完第1or2or3帧，output buffer尾部剩余的size < output buffer size /MaxStrmCnt即250K的时候，encoder也不会再在尾部接着放，而会从起始位置放。
4. 假设放完第1or2or3帧，output buffer尾部剩余size大于250K (假设是300K)，但是接下来encoder编出了一帧大于300K的帧，那就会出现pack=2的情况，即将这一帧300K放在尾部，剩余放不下的继续从起始位置放。

以上几种情况，当需要继续从起始位置放，但是用户一直没有取流或者取流慢导致第1帧还没有被取走，则encoder会停止编码，直到空间被腾出来。

5. 抓yuv图数据大小异常

- 现象描述

In some versions of sdk systems, the yuv image is captured through the api tool, and it is found that the size of the yuv image is twice the theoretical value, which cannot be viewed through the tool.

If the input resolution is 1920*1080, then if the captured YUV is YUV420, the image size should be 1920*1080*2Byte, but the actual size is

1920*1080*2*2Byte.

- Solution

The YUV data captured by these versions of the sdk is the YUV444 specification, so the image size is twice the theoretical value, but if the YUV valid data is arranged according to YUV444, that is, the invalid bits are filled with 0, which will cause the tool to view abnormally.

It needs to be converted by YUV444_conv.7z.

Open the execute file with notepad++, modify the yuv file to be converted and save it as shown, and then double-click execute. Then open it with 7yuv, pay attention to the format to choose YUV444.

